

objects concurrently.

Uncanny Vision is mimicking ways in which human vision and thoughts work through its IoT-Cloud platform for identifying anomalies rather than performing a direct image-to-image match for decision-making. Using this technology, it could be proved that such systems could be used to make our lives easier and secure.

Use cases in retail surveillance and analytics

Retail space has some interesting use cases for visual analytics based IoT systems. Visual data could be used to track movement of people across the display unit in the store and create heat-maps, paths or position-tracking. Path tracking could help a store manager to know the places where more people would possibly frequent and create a heat-map.

Apart from those, reactions and interests towards specific products could also be studied. This would sound advantageous for the retailer as he or she can position the products giving better visibility and thereby maximising probability of the customer buying the right product. The system could be taught to track multiple people and objects just in case there is a huge rush in the store.

Being informed

Uncanny Vision's IoT-Cloud infrastructure is wired to send out decisions and opinions in the form of notifications to a human with the available interfaces and media such as emails or push notifications on a



The Uncanny Vision team

handheld mobile device. A phenomenal amount of time is saved using Uncanny Vision as the end point device is intelligent enough to send out only processed decisions to the IoT-Cloud infrastructure, because 90 per cent of the time nothing interesting happens in front of a surveillance camera.

More than just sensing

In surveillance, most of the current 300 million cameras worldwide are blind and can only record video for post-mortem analysis when unexpected incidents happen. In some cases, there are basic analytics that are performed on servers using traditional image-processing algorithms like motion detection in high-security locations, people detection and so on.

Uncanny Vision's artificial intelligence/deep learning based algorithms can analyse videos a lot deeper and understand what actions humans perform, unexpected visual anomalies and the like.

In addition, Uncanny Vision's optimised surveillance software can run on high-performance cameras like Qualcomm's Snapdragon based wireless IP camera and carry out analytics on the device itself, making the system

more scalable.

Security and surveillance are incomplete without an intelligent system like Uncanny Vision in the days to come. There has been a lot of interest in this field and a multitude of applications are being developed to bring machines to a point of understanding intentions and act in unison with the biological world.

Predictability and machine learning are helping self-driven cars navigate and judge events on the road in real time. Streets could have night-vision-equipped cameras for pointing out crime and eventually eradicating such events.

In a hospital this system could monitor movement of a patient who needs long hours of observation and may need medical attention at will.

Uncanny Vision is more than a vision and can learn more than it can see. It is like having an eye at the back of the head with a mind of its own. Uncanny Vision provides surveillance solutions on a monthly subscription basis, depending on customer requirements. With availability of more data and fine tuning of convolutional neural network models, accuracy of artificial intelligence systems is expected to improve over time. **EFY**



Fig. 4: Retail analytics